

```

1  (*
2      CS51 Lab 7
3      Modules & Functors
4
5      A module for colors and color names
6
7  The representation for colors in this module is really obscure (and
8  arguably unnecessarily so). By the way, it also has some bugs so it
9  doesn't pass all the unit tests. No need to debug it though, or even
10 read it. You'll be replacing it wholesale with your own hopefully
11 simpler implementation, after constructing an appropriate signature
12 for the module in the file `color.mli`. *)
13
14 type color = int ;;
15
16 type color_name =
17   | Red
18   | Green
19   | Blue
20   | Orange
21   | Yellow
22   | Indigo
23   | Violet ;;
24
25 (* to_color r g b -- Returns the `color` corresponding to the RGB values
26    given by `r`, `g`, and `b` *)
27 let to_color (r : int) (g : int) (b : int) : color =
28   r lsl 0b10000 + g lsl 0b1000 + b ;;
29
30 (* red c / green c / blue c -- Returns the corresponding channel value
31    for the color `c` *)
32 let red_channel (c : color) : int =
33   c lsr 0b10000 ;;
34
35 let green_channel (c : color) : int =
36   (c lsr 0b1000) land 0b11111111 ;;
37
38 let blue_channel (c : color) : int =
39   c land 0b11111111 ;;
40
41 (* color_named name -- Returns the color corresponding to the color
42    `name` *)
43 let color_named (name : color_name) : color =
44   match name with
45   | Red -> 0b1111111100000000000000000000
46   | Green -> 0b11111111000000000
47   | Blue -> 0b11111111
48   | Orange -> 0b1111111110101101000000000
49   | Yellow -> 0b1111111111111110000000000
50   | Indigo -> 0b0101101100000000010000010
51   | Violet -> 0b1111000010000001011110100 ;;

```